

11 Supported protocols

This chapter is relevant for customers who want to configure the backup function in the Smart Home installation, i.e. read out the backup from an external unit. Modbus is an industry standard with which other devices can easily integrate and address the SAX Power Home (Plus). This means that systems from different manufacturers can be easily combined. Set values for active power and limit values for active power can be sent to the storage system. In addition, the state of charge (SOC) and the current output of the storage system can be read out. The protocols used by the storage system are described below.

11.1 Modbus TCP Control

Instructions for using the Modbus TCP interface: The storage unit offers a Modbus TCP interface via which communication is possible. This makes it possible not only to read out data, but also to write data to the storage unit.

Prerequisites

- Network access: The storage unit must be connected to the local network.
- IP address: The IP address of the storage device must be known.
- Modbus client software: A corresponding programme or tool for communication via Modbus TCP is required. (e.g. Modbus Poll, customised control software, etc.)

Establish connection

1. Network connection: Make sure that the storage unit is located in the same network segment as your computer or control system.
2. Configuring the Modbus client
 - Enter the IP address of the storage unit.
 - Select port 502 (standard port for Modbus TCP).
 - Configure the query parameters according to the following table.

Various slave IDs are available for control via Modbus TCP. The corresponding register lists apply depending on the ID used.

Note:



With Modbus TCP, a distinction is made between internal addresses and protocol addresses. If the read/write function is already specified in the client, the internal register address must be used. The register offset for function 03 (Read Holding Registers) is 40001, so to access the protocol address 40042, the internal register address 41 must be read.

11.2 Modbus TCP Basic Mode

Modbus Basic Mode is available to enable integration into energy management systems. This basically distinguishes between two operating modes, which can be selected via the customer service or the configuration software (parameter setting programme).

Smart Meter operating mode (RS485)

- Control of the storage unit is always zero control via the SAX Power Smart Meter.
- Values such as SOC, instantaneous power of the storage system and power at the grid connection point can be read out.
- Limit values for charging / discharging power can be written.

P setpoint operating mode (TCP)

- There is no regulation via the SAX Power Smart Meter. The storage system only works on the basis of the specified setpoints.
- Values such as SOC, instantaneous power of the storage system and power at the grid connection point can be read out.
- Setpoints must be specified to operate the storage unit.
- The power gradient of the setpoint control is 0.27%/s ex works.

The unit-ID in basic mode is always 0x40 (64).

Write commands must be repeated periodically. Repetition is recommended here at least every 5 minutes, but no more than every 5 seconds.

11.3 Slave-ID 64: Values – SAX Power Home (Plus)

Protocol address	Internal address	Designation	Unit	Unit-ID	Hint	Data typ	Access type
40042	41	Setpoint power P	Watt	64	(1)	uint16	Write
40043	42	Setpoint cos(Phi)	-	64	(1)	uint16	Write
40044	43	Power limit value for discharge	Watt	64	(2)	uint16	Write
40045	44	Power limit value for charge	Watt	64	(2)	uint16	Write
40046	45	Switching state of the storage unit	-	64	(3)	uint16	Read/Write
40047	46	SOC of the storage unit (measured value)	Prozent	64		uint16	Read
40048	47	Power P of the storage unit (measured value)	Watt	64		uint16	Read
40049	48	Power of the smartmeter	Watt	64		uint16	Read

(1) Note: Only available in P setpoint (TCP) operating mode.

(2) Note: 43, 44 are only available after activation.

(3) Note: There are two commands for Write: "Off" 01 and "On" 02.
There are three commands for Read: "Off" 01 "On" 02 and "Connected" 03.

11.4 Modbus-TCP Slave Extended Mode

This operating mode enables extended control of the storage unit and the reading of extended parameters from the connected smart meter.

Different slave IDs must be used here; these can be found in the corresponding table.

11.5 Slave-ID 40: Werte – SAX Power Home (Plus)

Protocol address	Internal address	Designation	Unit	Unit-ID	Hint	Data type	Access type
40071	70	SunSpec ID	–	40	–	uint16	Read
40072	71	SunSpec Length	–	40	–	uint16	Read
40073	72	Total phase currents	A	40	–	uint16	Read
40074	73	Current L1	A	40	–	uint16	Read
40075	74	Current L2	A	40	–	uint16	Read
40076	75	Current L3	A	40	–	uint16	Read
40077	76	Current scaling	sunssf	40	Scaling factor: -2	sunssf	Read
40078	77	N.A.	–	40	–	–	Read
40079	78	N.A.	–	40	–	–	Read
40080	79	N.A.	–	40	–	–	Read
40081	80	Voltage L1	V	40	–	uint16	Read
40082	81	Voltage L2	V	40	–	uint16	Read
40083	82	Voltage L3	V	40	–	uint16	Read
40084	83	Voltage scaling	–	40	Scaling factor: -1	–	Read
40085	84	Total power AC (active)	W	40	Scaling factor: 1	int16	Read
40086	85	Scaling total power (active)	sunssf	40	–	int16	Read
40087	86	Grid frequency	Hz	40	–	uint16	Read
40088	87	Grid frequency scaling	sunssf	40	Scaling factor: -1	sunssf	Read
40089	88	Total power AC (apparent)	VA	40	–	int16	Read
40090	89	Scaling total power (apparent)	sunssf	40	Scaling factor: 1	sunssf	Read
40091	90	Total power AC (reactive)	VAr	40	–	int16	Read
40092	91	Scaling total power (reactive)	sunssf	40	Scaling factor: 1	sunssf	Read
40093	92	Power factor	Per cent	40	–	int16	Read
40094	93	Power factor scaling	sunssf	40	Scaling factor: -1	sunssf	Read



Note: The scaling factor is to be regarded as an exponent of 10. A scaling factor of -1 therefore corresponds to a multiplier of 0.1.

11.6 Slave-ID 40: Values – Smartmeter

Protocol address	Internal address	Designation	Unit	Unit-ID	Hint	Data type	Access type
40095	94	N.A.	-	40	-	-	Read
40096	95	Energy fed in	kWh	40	-	uint16	Read
40097	96	Energy related	kWh	40	-	sunssf	Read
40098	97	Energy scaling	sunssf	40	Scaling factor: 1	-	Read
40099	98	Switching state of the memory	-	40	(1)	uint16	Read
40100	99	Current L1	A	40	Scaling factor: -2	int16	Read
40101	100	Current L2	A	40	Scaling factor: -2	int16	Read
40102	101	Current L3	A	40	Scaling factor: -2	int16	Read
40103	102	Active power L1	W	40	-	int16	Read
40104	103	Active power L12	W	40	-	int16	Read
40105	104	Active power L13	W	40	-	int16	Read
40106	105	Power scaling	sunssf	40	Scaling factor: 1	sunssf	Read
40107	106	Voltage L1	V	40	-	int16	Read
40108	107	Voltage L2	V	40	-	int16	Read
40109	108	Voltage L3	V	40	-	int16	Read
40110	109	Total power (active)	W	40	-	int16	Read



Note: The scaling factor is to be regarded as an exponent of 10. A scaling factor of -1 therefore corresponds to a multiplier of 0.1.

11.7 Slave-ID 123: Immediate Controls



Note: Will be supported with future software versions. (As of 03/25 not yet available)

Protocol address	Internal address	Designation	Unit	Unit-ID	Hint	Data type	Access type
40349	348	Power setpoint	%	123	Scaling factor: 100 -10000 ... +10000	int16	Read/Write
40353	352	Control mode (limiting)	-	123	(2)	int16	Read/Write

(1) Note: 1= OFF ; 2= ON ; 3= Connected ; 4= Standby

(2) Note: 0= Control via smart meter

1 = Setpoint control. If no new value is set for 5 minutes, control via smart meter (0) is automatically reactivated.



Note: Both registers must be written at the same time for successful setpoint control.

12 Decommissioning

To decommission, simply switch off the main switch. When switched off the storage unit is below the safety voltage (20V).

Attention:

Only switch off the storage unit for a longer period of time if the remaining charge level is at least 50%. This is necessary to protect the battery cells from deep discharge. Switch on the storage unit at least once **every 6 months** to prevent the battery cells from discharging too deeply.

13 Technical Data

13.1 SAX Power Home System data

Battery type	LiFePO4
Cell data	3.2V; 15Ah
Usable capacity	5.2kWh
Modularly expandable	up to 17.3kWh (3 storage units)
Nominal voltage	230V AC 1 phase
Rated power normal	4.6kVA
Rated current normal	20A
Storage weight	52kg
Storage dimension	530mm x 530mm x 250mm
Standby power / consumption	approx. 4W